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OUR RESEARCH QUESTION

Does coupling (CBO) still predict faults once class size is controlled?

Because a class tied to many others can break when any of them changes.

WHAT WE FOUND

CK metrics predict faulty classes, but **only until class size is controlled.**

After that, the evidence splits. Coupling (CBO) is the only metric that survives in any study.

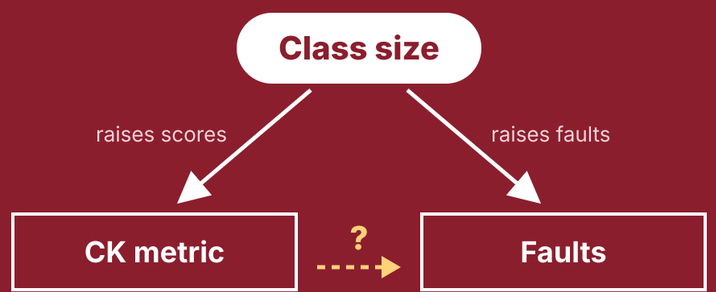


Fig. 1 In El Emam's study, adding class size made every metric's effect disappear.

Scan for the full evidence

Method details, key numbers, Q&A and references

The evidence trail

Each study answers a gap left by the one before

THE CLAIM → YES → NO → PARTLY → YES*

Chidamber & Kemerer 1994

- Proposes 6 OO design metrics: WMC, DIT, NOC, CBO, RFC, LCOM
- Grounded in Bunge's ontology; checked against Weyuker's 9 properties
- Collected 634 C++ + 1,459 Smalltalk classes at two firms

Pro: first theory-based OO metrics suite, with a tool to collect it

Con: never tested against faults; several metrics fail Property 6

Basili et al. 1996

- 8 student teams built the same video-rental system in C++
- 180 classes; faults found by an independent test team
- 5 of 6 metrics linked to faults; LCOM not, NOC reversed

Pro: same requirements for every team, so a fair comparison

Con: students and small systems; class size ignored

El Emam et al. 2001

- Ericsson C++ telecom framework: 174 classes, 70 faulty
- Fault = field failure reported by users (yes/no)
- WMC, CBO, RFC significant, until size is controlled: then none

Pro: first to test class size as a confounder

Con: one system and a small sample

Subramanyam & K. 2003

- Industry e-commerce suite: 405 C++ + 301 Java classes
- Defects from acceptance tests and customers; size controlled
- WMC, CBO raise defects in C++, not Java; deeper DIT, fewer defects

Pro: real industry data in two languages, size controlled

Con: only 3 metrics tested, one company

Gyimóthy et al. 2005

- Mozilla 1.6: 3,192 classes, bugs taken from Bugzilla
- Regression, decision trees and neural networks
- CBO best predictor; lines of code close; NOC useless

Pro: large real open-source system, tracked over 7 versions

Con: one project; *size not controlled

Where the studies disagree

Metrics that predict faults when size is ignored all fail in El Emam's size-controlled study.

* after controlling for class size
– not tested

	Basili 1996	El Emam 2001*	Subramanyam 2003*	Gyimóthy 2005
WMC	Yes	No	C++ only	Yes
DIT	Yes	No	Reversed	Weak
NOC	Reversed	–	–	No
CBO	Yes	No	C++ only	Best
RFC	Yes	No	–	Yes
LCOM	No	No	–	Yes

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